

14 January 2025

To Whom It May Concern

Letter of Support for Mr Manas Mahale

Sir or Madam,

I am writing to recommend Manas Mahale, who has worked with me since 2022 in various capacities; initially with my cheminformatics/ML in life sciences research group at the University of Cambridge, subsequently as an AI and Cheminformatics Intern at Pangea Botanica/Pangea Bio in Berlin (full time for 9 months, 2022/2023), and currently as an affiliate group member at Universitatea Babeş-Bolyai (UBB). During this time, Manas has contributed significantly to several projects and consistently demonstrated a strong command of computational and cheminformatics techniques.

At Cambridge as well as Pangea Bio, Manas worked on molecular similarity prediction using tandem mass spectra, where he focused on the impact of positional embeddings on model performance. He also explored the application of masked language models for small molecular library generation in medicinal chemistry. These projects required a strong foundation in computational methods and attention to detail, both of which Manas handled effectively. In particular, Manas' capabilities in both being able to understand life science data (e.g. chemical structures) and his excellent ML background were the basis for that.

During his time at Pangea Bio, Manas developed LILA, a ligand-target prediction suite aimed at streamlining drug discovery workflows. He improved the performance of the MS2DeepScore model for predicting natural product structural similarity from MS/MS spectra and established workflows for pharmacokinetic prediction models, deploying them effectively on AWS. He also automated processes for protein alignment and homology modeling using Schrödinger's Python API, improving accuracy and efficiency. These contributions highlighted his ability to deliver practical and impactful solutions in computational drug discovery. Furthermore, they underline his abilities also across further life science domains (e.g. pharmacokinetics), as well as technically (Cloud Engineering).

Manas's more recent work at UBB has involved developing ligand-target prediction and mode-of-action analysis models to enhance phenotypic screening of small molecules, together with current PhD students in the group. His ongoing research focuses on foundational models for single-cell perturbation data, which is rather heavy on the ML domain, and also requires a good understanding of the biological type of data supplied to the models. In addition, his development of cheminformatics tools such as DILIPredictor for drug-induced liver injury prediction and Analogue Split, a data-

splitting method for activity cliffs, reflects his focus on building reliable, scientifically rigorous methods to solve specific challenges in the field.

From the personality side, Manas approaches his work with focus, precision, and a structured manner. He reliably delivers scientific outcomes related to all of the diverse disciplines mentioned above; and he also communicates in a very constructive and personable manner, both with other group members as well as with myself, and people outside the respective institution he was working in, in case of joint project.

His programming expertise, while at the same time being able to understand the life science behind data he was modelling, has made him a tremendously valuable contributor to the development of reproducible tools like PKSmart and MolecularNetwork, which are now available for broader scientific use. Manas' work has also resulted in a series of joint publications already; some are available at <https://scholar.google.com/citations?user=blcV6MQAAAAJ&hl=en&oi=ao>, but several more publications and preprints have now been submitted in peer-reviewed journals, which I think is an excellent result for Manas' career stage.

Given the above, I can wholeheartedly recommend Manas for positions that are situated at the interface of life sciences and life science data, and data analytics/AI/ML, be they more related to research (such as PhD positions), as well as roles in related companies (due to his ability to also deliver in a company context) His technical skills, problem-solving abilities, and professional approach to projects make him a reliable and capable contributor.

In case of further questions feel free to contact me.

Sincerely,



Professor Andreas Bender
Visiting Professor of Molecular Informatics
Department of Chemistry, University of Cambridge